COLLECTION OF CTL DECISIONS

Standard: IEC 61010-1:1990+A1:1992+A2:1995	**Sub clause:** 14	**Sheet n. 310** **Page 1(1)**
Subject: Components	**Key words:** - Transient - Suppressor - Mains	**Decision taken by ETF3 and confirmed by CTL at its 38th meeting, in Toronto**

Question:
Is the use of transient suppressor devices such as varistors accepted in equipment and, if so, under what conditions

Decision:
Varistors are accepted in the following conditions:

1 Between Live and neutral conductors after the mains fuse.
No component approval demanded.

2 Between Live and neutral conductors, before the mains fuse.
Only accepted if the component is approved to an international standard (IEC 61051 or CECC 42200)

3 Between Live and Earth conductors (before or after the mains fuse)

a) pluggable equipment
devices are not accepted

b) permanently installed equipment
devices are accepted as in 1 and 2 above.

Note: Components shall be physically mounted in such a way that no hazard can arise in the event of the device failing catastrophically.

Explanation:

Varistors are semi-conductor devices and therefore, cannot be treated as high integrity devices and must therefore, always have additional protection.

Varistors in the equipment or installation coupled with safe construction and approvals meet these requirements except where the disconnection of earth is possible as a single fault. In this case, the slow deterioration of the device with time, allowing an increase in earth leakage current, is not acceptable and the devices are, therefore, not allowed.